WEEK 2 POWER BI PROJECT REPORT ：DATA MODELING, DAX & SALES DASHBOARD

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Week 1 Task Recap:

In Week 1, I completed the Superstore Sales Data Import, Power Query and Data Cleaning task. I successfully imported the Superstore dataset into Power BI, inspected the data, checked data types and data quality, performed the required data cleaning and transformations, and created the required date-related columns such as Year, Quarter, Month, Day and Day of Week. The cleaned and transformed dataset was then loaded into Power BI and prepared for data modeling and dashboard task.

Introduction:

In Week 2, the cleaned Superstore dataset prepared during Week 1 was used to develop a structured data model and an interactive sales dashboard using Microsoft Power BI. The main focus of this week was to organize the data using a Star Schema, create important business calculations using DAX, and present the results through interactive visualizations.

The objective was to transform the cleaned transactional data into a dashboard that provides a clear overview of sales, profit**,** orders, quantity, category performance, monthly trends, and regional profitability. The Week 2 task also required the use of KPI cards, charts, a decomposition tree, and slicers for interactive analysis.

Objectives

The main objectives of Week 2 were:

- To create a proper Star Schema data model.
- To divide the dataset into fact and dimension tables.

- To establish relationships between the tables.
- To create a dedicated _Measures table.
- To create important DAX measures for business analysis.
- To develop KPI cards for quick performance monitoring.
- To analyze sales by category and month.
- To analyze profitability by region.
- To create a Decomposition Tree for detailed profit analysis.
- To add slicers and interactive dashboard features.
- To develop a clear and user-friendly Superstore Sales Insights Dashboard.

Creating Product and Customer Dimension Tables

To convert the cleaned Superstore dataset into a Star Schema, two additional dimension tables—Product and Customer—were created from the FactSales table using Power Query.

First, the FactSales table was duplicated and the duplicate was renamed Product. Only the product-related columns such as Product ID, Product Name, Category, and Sub-Category were retained, and duplicate Product IDs were removed. This allowed each product to appear only once in the Product dimension.

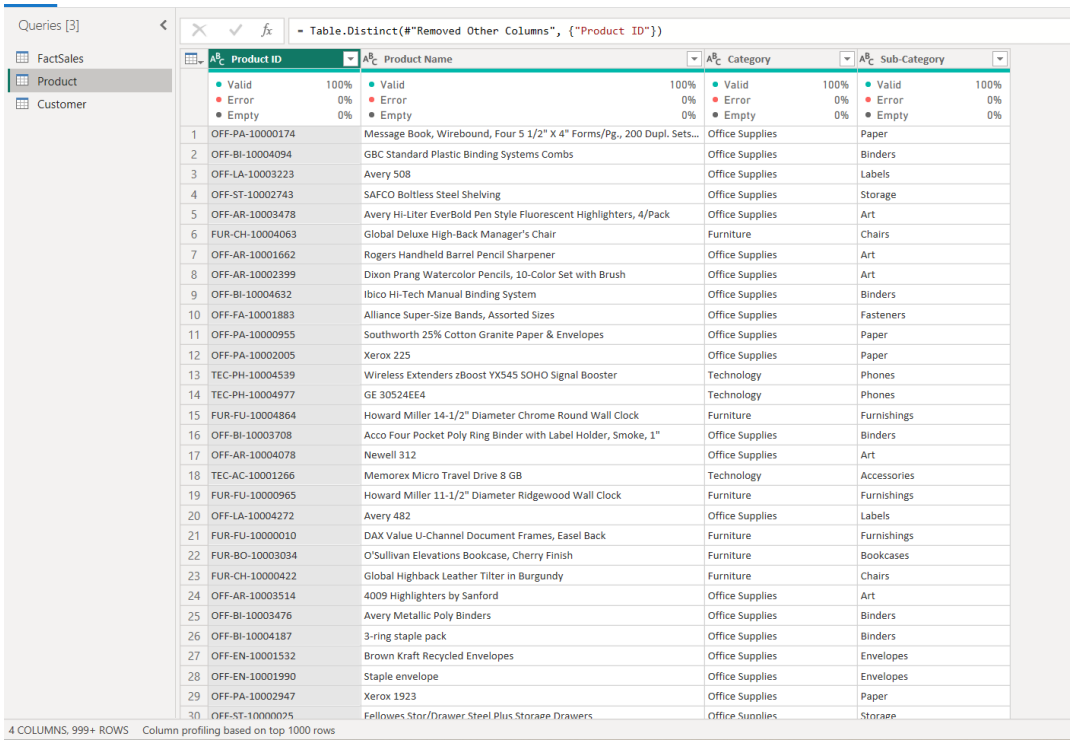
The FactSales table was then duplicated again to create the Customer table. Customer-related columns were retained, and the table was connected to FactSales using Customer ID.

Finally, one-to-many relationships were established between these dimension tables and the FactSales table:

- Product (1) → FactSales (*)
- Customer (1) → FactSales (*)

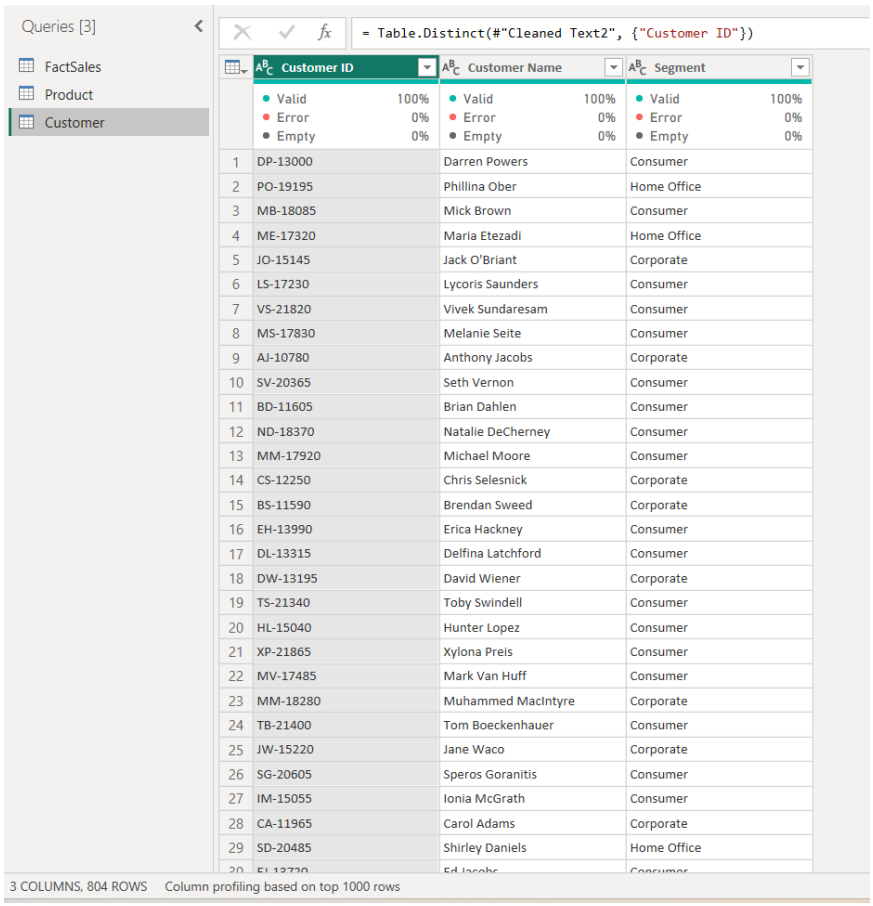
This separation of product and customer information helped organize the dataset into a proper Star Schema and made it easier to perform filtering and analysis in Power BI.

Product Dimension:



	Product ID	Product Name	Category	Sub-Category
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
1	OFF-PA-10000174	Message Book, Wirebound, Four 5 1/2" X 4" Forms/Pg., 200 Dupl. Sets...	Office Supplies	Paper
2	OFF-BI-10004094	GBC Standard Plastic Binding Systems Combs	Office Supplies	Binders
3	OFF-LA-10003223	Avery 508	Office Supplies	Labels
4	OFF-ST-10002743	SAFCO Boltless Steel Shelving	Office Supplies	Storage
5	OFF-AR-10003478	Avery Hi-Liter EverBold Pen Style Fluorescent Highlighters, 4/Pack	Office Supplies	Art
6	FUR-CH-10004063	Global Deluxe High-Back Manager's Chair	Furniture	Chairs
7	OFF-AR-10001662	Rogers Handheld Barrel Pencil Sharpener	Office Supplies	Art
8	OFF-AR-10002399	Dixon Prang Watercolor Pencils, 10-Color Set with Brush	Office Supplies	Art
9	OFF-BI-10004632	Ibico Hi-Tech Manual Binding System	Office Supplies	Binders
10	OFF-FA-10001883	Alliance Super-Size Bands, Assorted Sizes	Office Supplies	Fasteners
11	OFF-PA-10000955	Southworth 25% Cotton Granite Paper & Envelopes	Office Supplies	Paper
12	OFF-PA-10002005	Xerox 225	Office Supplies	Paper
13	TEC-PH-10004539	Wireless Extenders zBoost YX545 SOHO Signal Booster	Technology	Phones
14	TEC-PH-10004977	GE 30524EE4	Technology	Phones
15	FUR-FU-10004864	Howard Miller 14-1/2" Diameter Chrome Round Wall Clock	Furniture	Furnishings
16	OFF-BI-10003708	Acco Four Pocket Poly Ring Binder with Label Holder, Smoke, 1"	Office Supplies	Binders
17	OFF-AR-10004078	Newell 312	Office Supplies	Art
18	TEC-AC-10001266	Memorex Micro Travel Drive 8 GB	Technology	Accessories
19	FUR-FU-10000965	Howard Miller 11-1/2" Diameter Ridgewood Wall Clock	Furniture	Furnishings
20	OFF-LA-10004272	Avery 482	Office Supplies	Labels
21	FUR-FU-10000010	DAX Value U-Channel Document Frames, Easel Back	Furniture	Furnishings
22	FUR-BO-10003034	O'Sullivan Elevations Bookcase, Cherry Finish	Furniture	Bookcases
23	FUR-CH-10000422	Global Highback Leather Tilter in Burgundy	Furniture	Chairs
24	OFF-AR-10003514	4009 Highlighters by Sanford	Office Supplies	Art
25	OFF-BI-10003476	Avery Metallic Poly Binders	Office Supplies	Binders
26	OFF-BI-10004187	3-ring staple pack	Office Supplies	Binders
27	OFF-EN-10001532	Brown Kraft Recycled Envelopes	Office Supplies	Envelopes
28	OFF-EN-10001990	Staple envelope	Office Supplies	Envelopes
29	OFF-PA-10002947	Xerox 1923	Office Supplies	Paper
30	OFF-ST-10000025	Fellowes Stor/Drawer Steel Plus Storage Drawers	Office Supplies	Storage

Customer Dimension:



	Customer ID	Customer Name	Segment
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
1	DP-13000	Darren Powers	Consumer
2	PO-19195	Phillina Ober	Home Office
3	MB-18085	Mick Brown	Consumer
4	ME-17320	Maria Etezadi	Home Office
5	JO-15145	Jack O'Briant	Corporate
6	LS-17230	Lycoris Saunders	Consumer
7	VS-21820	Vivek Sundaresam	Consumer
8	MS-17830	Melanie Seite	Consumer
9	AJ-10780	Anthony Jacobs	Corporate
10	SV-20365	Seth Vernon	Consumer
11	BD-11605	Brian Dahlen	Consumer
12	ND-18370	Natalie DeCherney	Consumer
13	MM-17920	Michael Moore	Consumer
14	CS-12250	Chris Selesnick	Corporate
15	BS-11590	Brendan Sweed	Corporate
16	EH-13990	Erica Hackney	Consumer
17	DL-13315	Delfina Latchford	Consumer
18	DW-13195	David Wiener	Corporate
19	TS-21340	Toby Swindell	Consumer
20	HL-15040	Hunter Lopez	Consumer
21	XP-21865	Xylona Preis	Consumer
22	MV-17485	Mark Van Huff	Consumer
23	MM-18280	Muhammed MacIntyre	Corporate
24	TB-21400	Tom Boeckenhauer	Consumer
25	JW-15220	Jane Waco	Corporate
26	SG-20605	Speros Goranitis	Consumer
27	IM-15055	Ionia McGrath	Consumer
28	CA-11965	Carol Adams	Corporate
29	SD-20485	Shirley Daniels	Home Office
30	EJ-13720	Ed Jacobs	Consumer

Data Modeling

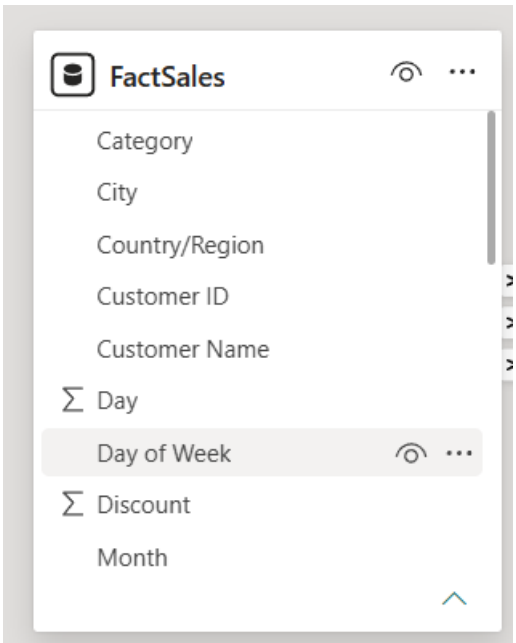
The first major task in Week 2 was to organize the cleaned dataset into a suitable data model.

Instead of keeping all information in one large table, the data was structured into separate tables to create a Star Schema.

FactSales

The FactSales table was used as the central fact table. It contains the transactional sales information, including fields related to:

- Order ID
- Customer ID
- Product ID
- Order Date
- Sales
- Profit
- Quantity
- Category
- Region
- Sub-Category

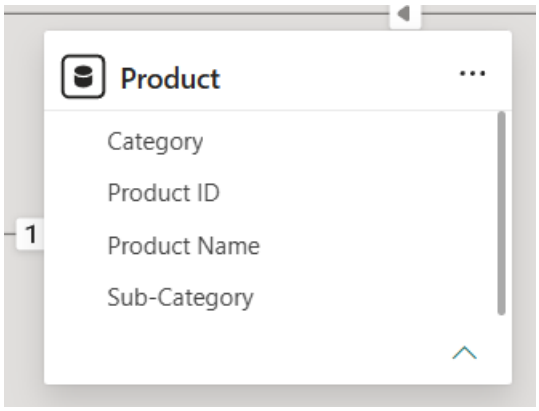


Product Table

A separate Product dimension table was created containing product-related information such as:

- Product ID
- Product Name
- Category
- Sub-Category

Duplicate Product IDs were removed so that the Product table could correctly function as a dimension table.

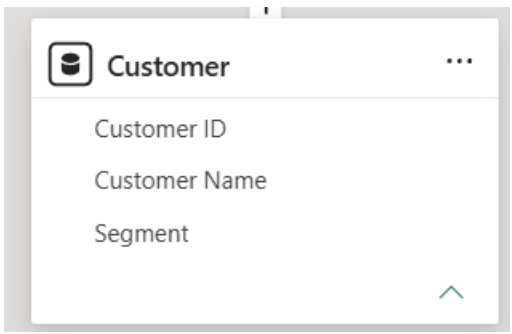


Customer Table

A separate Customer dimension was created to store customer-related information such as:

- Customer ID
- Customer Name
- Segment

The relationship was configured so that one customer can have many sales transactions.

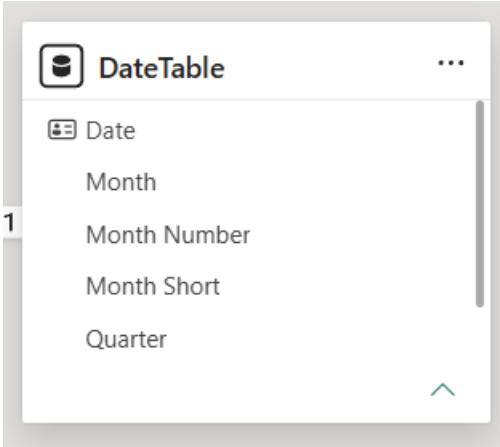


DateTable

A separate DateTable was created to support time-based analysis. It contains:

- Date
- Year
- Quarter
- Month
- Month Number
- Month Short

The DateTable was marked as the official date table in Power BI and connected to the Order Date field in FactSales.



Establishing Relationships

Relationships were created between the fact and dimension tables to form the Star Schema.

The main relationships were:

- Product → FactSales
- Customer → FactSales
- DateTable → FactSales

The dimension tables were placed on the one side (1) and FactSales on the many side (*).

This allows filters from the dimension tables to correctly affect the transactional data.

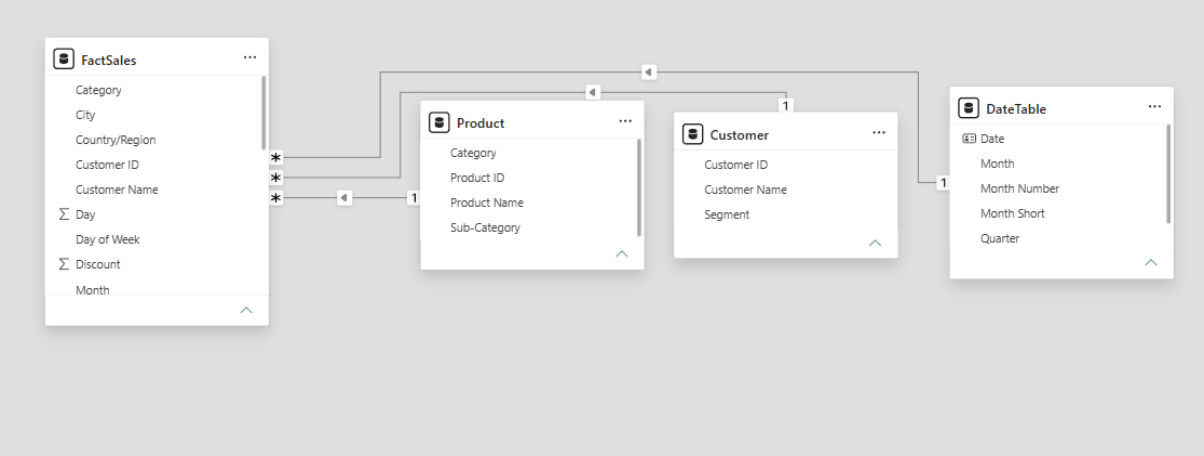
Data Model Structure

Product (1) → FactSales (*)

Customer (1) → FactSales (*)

DateTable (1) → FactSales (*)

The relationships were configured as active relationships with single cross-filter direction.



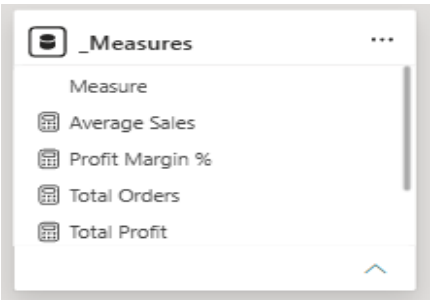
Dedicated Measures Table

A dedicated table named _Measures was created to store all the DAX calculations in one place.

This keeps the data model organized and makes the measures easier to locate and manage.

The following six measures were created:

1. Total Sales
2. Total Profit
3. Total Orders
4. Total Quantity
5. Average Sales
6. Profit Margin %



DAX Measures

Total Sales

Total Sales = SUM(FactSales[Sales])

This measure calculates the total sales generated from all transactions.

Result: Approximately **2.33M**

Total Profit

Total Profit = SUM(FactSales[Profit])

This measure calculates the total profit generated from the sales transactions.

Result: Approximately **292.13K**

Total Orders

Total Orders = DISTINCTCOUNT(FactSales[Order ID])

This measure counts the number of unique orders in the dataset.

Result: Approximately **5K**

Total Quantity

Total Quantity = SUM(FactSales[Quantity])

This measure calculates the total number of products sold.

Result: Approximately **39K**

Average Sales

Average Sales = AVERAGE(FactSales[Sales])

This measure calculates the average sales value across the transactions.

Result: Approximately **228.3**

Profit Margin %

Profit Margin % =

```
DIVIDE(
  SUM(FactSales[Profit]),
  SUM(FactSales[Sales]),
  0
)
```

This measure calculates the percentage of sales that is retained as profit.

Result: Approximately **12.6%**

KPI Cards:

Six KPI cards were created at the top of the dashboard to provide a quick overview of the business performance.

KPI	Value
Total Sales	2.33M
Total Profit	292.13K
Total Orders	5K
Total Quantity	39K
Average Sales	228.3
Profit Margin %	12.6%

These cards allow users to understand the overall performance without having to examine individual charts.

Creating Monthly Sales Trend

A line chart was created to show how sales changed over time. The Month field from the DateTable was used along the axis, while Total Sales was used as the value.

The Month Number was used for proper chronological sorting so that the months appear in the correct order.

Creating Regional Profitability Chart

A chart was added to compare profitability across different regions. Region was used as the category and Total Profit was used as the value.

This makes it easier to identify regions generating higher or lower profits.

Adding Decomposition Tree

A Decomposition Tree was included to provide deeper analysis of profit. Total Profi**t** was selected under *Analyze*, while Category, Region, and Sub-Category were added under *Explain By*.

This allows the user to interactively break down total profit and understand which categories, regions, and sub-categories contribute to the result.

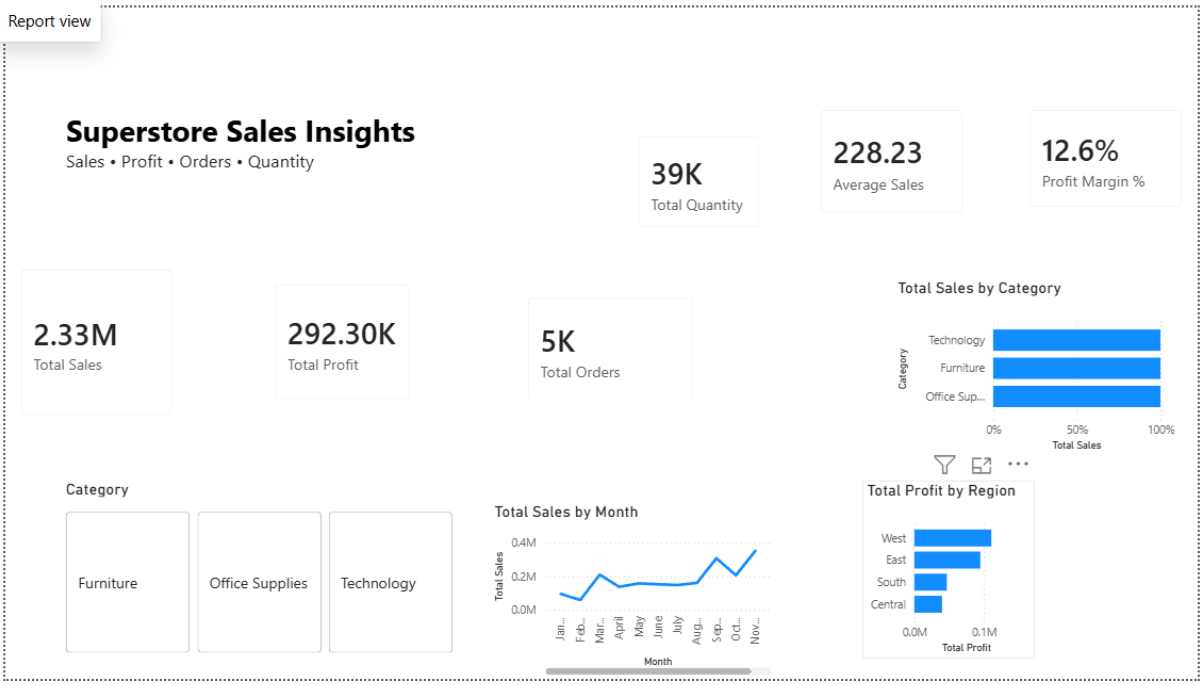
Adding Interactive Slicer

A Category slicer was added to make the dashboard interactive. Users can select a particular category and the other visuals on the dashboard automatically update according to the selected category.

Dashboard Layout and Design

Finally, all visuals were arranged into a single dashboard with the title "Superstore Sales Insights" and the subtitle "Sales • Profit • Orders • Quantity."

The KPI cards were positioned at the top for quick reference, while the charts and Decomposition Tree were arranged below them. The dashboard was kept simple and organized so that important information could be understood easily.



Dashboard Outcome:

The completed dashboard provides an interactive view of Superstore sales performance. It allows users to quickly monitor important KPIs, compare product categories, observe monthly sales trends, analyze regional profitability, and drill down into profit using the Decomposition Tree. The use of DAX measures, relationships, slicers, and interactive visuals makes the dashboard useful for business performance analysis.

Key Results:

The dashboard generated the following key performance indicators:

- **Total Sales:** 2.33M
- **Total Profit:** 292.13K
- **Total Orders:** 5K
- **Total Quantity:** 39K
- **Average Sales:** 228.3

- **Profit Margin:** 12.6%

The visualizations make it easier to compare categories, monitor monthly sales, analyze regional profitability and investigate profit using the Decomposition Tree.

Conclusion:

Week 2 successfully transformed the cleaned Superstore dataset into a structured Power BI data model and interactive dashboard. The implementation of a Star Schema improved the organization of the dataset, while DAX measures provided the calculations required for business analysis.

The **Superstore Sales Insights Dashboard** brings together key performance indicators and interactive visualizations to provide a clear understanding of sales and profitability. The project demonstrates the practical application of **data modeling, DAX, Power Query and Power BI visualization techniques**.